Week 3 Worksheet (Solutions)

COMP2048

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Review of Definitions and Notation

Concatenation. For strings x and y, xy denotes the string formed by appending y to x. We have $x\varepsilon = \varepsilon x = x$.

Concatenation of Languages. For languages $L_1, L_2 \subseteq \Sigma^*$, the concatenation of L_1 and L_2 is

$$L_1L_2 = \{\, xy \mid x \in L_1,\ y \in L_2 \,\}.$$

That is, L_1L_2 contains all strings formed by taking a string from L_1 followed by a string from L_2.

Kleene Star. For a language $L \subseteq \Sigma^*$, the Kleene star of L is

$$L^* = \{\varepsilon\} \cup L \cup LL \cup LLL \cup \cdots .$$

Equivalently,

$$L^* = \{\, w_1w_2 \cdots w_k \mid k \geq 0,\ w_i \in L \,\}.$$

It contains all strings obtained by concatenating zero or more strings from L.

Language of a DFA. For a DFA M, the language recognized by M is

$$L(M) = \{\, w \in \Sigma^* \mid M \text{ accepts } w \,\}.$$

Regular Expression. A regular expression (regex) describes a regular language using:

- $\emptyset$, ε, and symbols from Σ
- union ($R \cup S$)
- concatenation (RS)
- star (R^*) (i.e., "zero or more")

Order of operations (i) parentheses, (ii) star, (iii) concatenation, (iv) union.

NFA (Nondeterministic Finite Automaton). Like a DFA but $\delta : Q \times \Sigma_\varepsilon \to \mathcal{P}(Q)$, where $\Sigma_\varepsilon = \Sigma \cup \{\varepsilon\}$ and $\mathcal{P}(Q)$ is the power set of Q (i.e., set of all subsets of Q). An NFA accepts if *any* computation path leads to an accept state. Every NFA has an equivalent DFA.

Regular Language. A language recognized by some DFA (equivalently, by some NFA, or described by some regular expression).

Task 1. *Regular Languages*

Prove that the following languages over the alphabet $\Sigma = \{0, 1\}$ are regular:

(a) $L_a = \emptyset$

(b) $L_b = \Sigma^*$

(c) $L_c = \{\varepsilon\}$

(d) $L_d = \{1111\}$

Solution.

A language is by definition regular it is recognized by some DFA. Draw these diagrams for them:

(a) DFA with one non-accepting state that loops on both 0 and 1 accepts no strings.

(b) DFA with one accepting state that loops on both 0 and 1 accepts every string over Σ.

(c) DFA with two states whose start state is accepting, and any input symbol sends the machine to a non-accepting trap state.

(d) DFA that reads exactly four 1's (i.e., 5 states) and accepts only if the input ends there; any 0 at any state leads to a non-accepting trap state. The accepting state also transitions to the trap state if it reads a 1.

Task 2. *Kleene Star and Concatenation of Languages*

Let $L = \{0, 11\}$ over the alphabet $\Sigma = \{0, 1\}$.

(a) List all strings in LL (sometimes denoted by L^2).

(b) Give three strings in L^* of length at least 3.

(c) Give one string in $\{0, 1\}^*$ that is *not* in L^*.

Solution.

Strings in LL is formed by concatenating one string from L with another string from L.

(a)
$$LL = \{00,\ 011,\ 110,\ 1111\}.$$

(b) Since
$$L^* = \{\varepsilon\} \cup L \cup LL \cup LLL \cup \cdots,$$
it contains all strings formed by concatenating zero or more copies of 0 and 11.

Examples of strings in L^* of length at least 3 are:
$$000, \qquad 011, \qquad 110, \qquad 1111, \qquad \ldots$$

(c) Example of strings in $\{0,1\}^*$ that are not in L^* is
$$1, \qquad 111, \qquad \ldots$$
This is because strings in L^* are built from blocks 0 and 11, so a single 1 cannot be formed.

Task 3. *Strings Generated by a Regular Expression*

Let $\Sigma = \{a, b\}$ and consider the lnaguage generated by the following regular expression
$$ab(a \cup bab)^* \cup bbb$$

(a) Give five strings that belong to the language.

(b) Give five strings that do not belong to the language.

(c) Is the string $abbbb$ in this language?

Solution.

Remind them about the order of operations (see the yellow review box). The regular expression
$$ab(a \cup bab)^* \cup bbb$$
describes all strings that are either:

- exactly bbb, or
- start with ab and are followed by zero or more copies of a or bab.

(a) Five strings in the language are:
$$ab, \qquad aba, \qquad abbab, \qquad abaabab, \qquad bbb.$$

(b) Five strings not in the language are:

$$a, \qquad b, \qquad abb, \qquad abab, \qquad baab.$$

(c) No, the string $abbbb$ is not in the language.

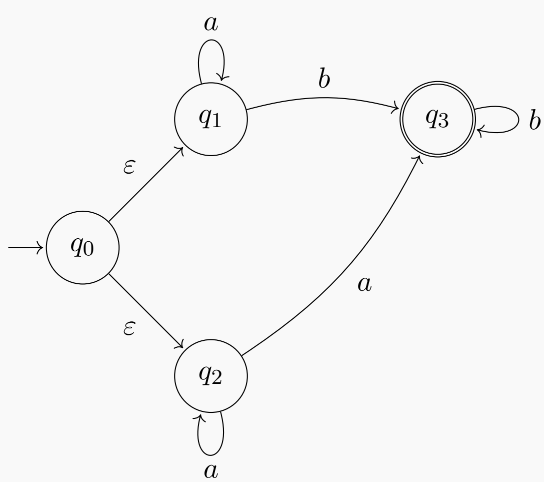
After the initial ab, the remaining part would have to be formed from blocks a and bab. But

$$abbbb = ab \circ bb,$$

and bb cannot be written as a concatenation of a's and bab's. So $abbbb \notin L$.

Task 4. *NFA*

Consider the following NFA over $\Sigma = \{a, b\}$.



(a) What states could the NFA be in after reading the empty string ε?

(b) What states could the NFA be in after reading the string a?

(c) What states could the NFA be in after reading the string aa?

(d) Determine whether the NFA accepts each of the following strings:

$$b, \quad ab, \quad aab, \quad aaa, \quad abb.$$

(e) Give a description of the language recognized by this NFA.

Solution.

Because of the ε-transitions from q_0, before reading any input the NFA may move to q_1 or q_2.

(a) After reading ε, the NFA could be in

$$\{q_0, q_1, q_2\}.$$

(b) After reading a:

- take an ε transition to q_1 and from q_1, reading a stays in q_1;
- take an ε transition to q_2 and from q_2, reading a can go to q_2 or q_3.

So the possible states are

$$\{q_1, q_2, q_3\}.$$

(c) We already worked out the set of states the machine can be in after reading the first a. Now after the reading the second a (in aa):

- from q_1, the second a keeps the machine in q_1;
- from q_2, the second a can keep it in q_2 or send it to q_3;
- from q_3, there is no transition on a, so that path dies;

Hence the possible states are

$$\{q_1, q_2, q_3\}.$$

(d) We test each string.

- b: accepted, since $q_0 \xrightarrow{\varepsilon} q_1 \xrightarrow{b} q_3$.
- ab: accepted, since $q_0 \xrightarrow{\varepsilon} q_1 \xrightarrow{a} q_1 \xrightarrow{b} q_3$.
- aab: accepted, since $q_0 \xrightarrow{\varepsilon} q_1 \xrightarrow{a} q_1 \xrightarrow{a} q_1 \xrightarrow{b} q_3$.
- aaa: accepted, since from q_2 one of the a's can move to q_3 after the whole input is consumed. For example:

$$q_0 \xrightarrow{\varepsilon} q_2 \xrightarrow{a} q_2 \xrightarrow{a} q_2 \xrightarrow{a} q_3.$$

- abb: accepted, since $q_0 \xrightarrow{\varepsilon} q_1 \xrightarrow{a} q_1 \xrightarrow{b} q_3 \xrightarrow{b} q_3$.

So all five strings are accepted.

(e) The language consists of:

- all strings of one or more a's, since from q_2 the machine may jump to q_3 on the last a;
- all strings of the form $a^n b^m$ where $n \geq 0$ and $m \geq 1$, since from q_1 the machine may read any number of a's, then one b to enter q_3, and then any number of additional b's.

So one description is:

$$L = \{a^n \mid n \geq 1\} \cup \{a^n b^m \mid n \geq 0,\ m \geq 1\}.$$